SI 699 - Milestone 3: Design Proposal

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Formative Research

As part of my formative research I carried out a survey in order to find out how women experience and cope with menstrual pain, where they go for information, and what relief methods they find useful. The survey had three main aims. The first of these was to find out which sources women use to get information on menstrual symptoms, how often they use them, and which ones they consult first and which ones they use when they need further information. The second aim was to investigate how frequently women carry on with or minimize their period pain, how often they delay getting help, and how comfortable and confident they are in managing their symptoms. Finally, I wanted to know how long women wait before reacting to their pain, whether by applying a relief method, stopping an activity, asking for help, or looking for information and what factors cause these delays. The survey was sent out via three different channels: the UMSI chatrooms, the women-led student groups that I am involved with, and friends from different parts of the world.

The survey made several interesting observations concerning the way in which women search for information and cope with menstrual pain. It appeared that the participants generally knew which sources were most effective for them when looking for information. Instead of using AI agents or chatbots, they more often turned to internet search engines and social media. A number of the respondents said that they had been suffering from period pain during most of their menstrual cycles in the past three months. One of the main reasons why the participants searched for information was to find out if their pain was normal. When asked what relief methods they would like to find out more about, they showed interest in finding out which foods might help to alleviate the pain and in discovering what they could do when experiencing pain at class or work. The responses also indicated that the participants did not hold strongly formed views regarding the trustworthiness of information on menstrual pain, the level of trust generally being neutral. Yet the most important factor when searching for information was discovering practical ways of reducing or stopping the pain.

After the survey, I carried out some collaborative user interviews in order to look more closely at experiences of menstrual pain. During these interviews I aimed to find out how women go from noticing a symptom to interpreting it, how they assess different kinds of information sources, and what affects their decision to get professional care. I also wished to investigate the informational, emotional, financial and logistical obstacles which might stop someone from seeking help. The participants were selected from those who had said in the survey that they were willing or might be willing to take part in an interview. Four women took part in the interviews, which included talks about their experiences of menstrual pain, the period kits they use, their preferred methods of relief, and what they would like others to understand about suffering period pain. The interviews were structured to help the participants think about their experiences in a comfortable and collaborative way.

A key early result from the interviews was that managing menstrual pain involves not only finding an effective way to relieve it but also having the freedom, the necessary resources, and support in order to deal with the pain when it arises. The participants gave a variety of accounts of their pain experiences, explaining that even familiar pain could cause frustration, while unexpected symptoms might lead to confusion or uncertainty. It also came out in the interviews that rest can act as a method of pain management although it is not always available. Since they had different academic, professional, and personal obligations, some of the participants had more flexibility than others to take account of their pain. This indicates that the management of menstrual pain is greatly affected by a person's surroundings and situation.

We also noticed that the participants would sometimes carry on with relief methods even though they did not think these were especially effective; the choices they made were shaped by their past experiences, the symptoms they were experiencing at the time, and the resources they had available to them then. Moreover, the participants showed signs of having established preparedness habits, for example by keeping necessary menstrual products or relief items to hand. Although such behaviours could be taken into account when designing a physical or hybrid type of intervention, the interviews did not show that there was a clear requirement for a completely new period kit or pouch, since the participants already had their own ways of managing their symptoms.

The interviews also highlighted the emotional and social dimensions of managing menstrual pain. The participants explained that emotional comfort, self-compassion, and having a supportive environment could make painful days easier to handle. It didn't have to involve getting direct physical relief in order to feel supported; instead, being acknowledged, having patience, receiving practical help, and being able to communicate their needs were also important. Moreover, if people experienced unexpected or unfamiliar symptoms they might wonder whether such an experience was normal, though further research would be required in order to understand how the participants deal with this uncertainty and decide whether to get medical help.

From these initial findings, I spotted four design opportunities to look into during the ideation phase. The first of these is about assisting people in managing their menstrual pain when they aren't able to stop what they're doing, for example in class or at work. The second looks at how an intervention could give emotional comfort even if it doesn't commit to providing a reduction in physical pain. The third looks at making it easier for people to access the relief methods that they already prefer, whether they are at home or away from home. Lastly, the fourth focuses on promoting communication, self-compassion, and recovery without imposing an extra cognitive load during an experience that is already unpleasant.

The formative research has enabled me to gain a more comprehensive understanding of how to manage menstrual pain beyond just the physical symptoms. Although the survey gave me some initial information about women's information-seeking habits and their preferences for relief, it was the interviews that showed me how personal circumstances, emotional needs, and access to support affect the way women deal with pain. These findings will inform the next phase of

ideation, focusing on creating interventions that are flexible, accessible, supportive, and take into account people's current routines and coping strategies.

Design Proposal

Design Implications

Based on the findings from the formative research, I identified several design implications that could guide the development of my intervention.

- From the interviews and survey, the following design implications were devised:
- Don't promise relief on the worst days.
- Design for "my" toolkit, not one recommendation.
- Organize relief by situation first (class, commute, work), then by technique.
- Make relief usable without stopping daily activities.
- Treat permission and support as part of pain management.
- Keep pain relief and period supplies conceptually separate.
- Add a visible "when to get this checked" path as a guardrail, not a main feature.
- A simple "did that help?" loop is a candidate.

One of the most important considerations was recognizing that an intervention should not promise complete pain relief, especially during particularly painful menstrual cycles. The interviews revealed that participants often relied on familiar relief techniques even when they were not always effective. This suggested that the design should focus on supporting users through their experiences rather than guaranteeing that their pain will disappear. Another important implication was the need for personalization. Rather than creating an application that recommends the same remedies to everyone, I wanted to design around each user's existing toolkit of strategies, preferences, and available resources.

The research also highlighted the importance of designing for different situations. Participants described how their ability to respond to menstrual pain depended on whether they were at home, attending classes, working, or managing other responsibilities. Rest, for example, was a preferred coping strategy for some participants, but it was not always possible because of their schedules. This led me to consider organizing relief techniques based on the user's current situation before suggesting specific remedies. The intervention should prioritize strategies that can be accessed and used without requiring someone to completely stop their daily activities. Additionally, emotional support, encouragement, and the ability to communicate personal needs emerged as important aspects of pain management. Therefore, the design should consider not only physical relief but also how to create a supportive environment where users feel comfortable acknowledging and responding to their pain.

Other design considerations included keeping menstrual supplies and pain-relief techniques conceptually separate, since being prepared with period products does not necessarily mean being prepared to manage pain. I also identified an opportunity to incorporate a visible pathway for understanding when symptoms may require medical attention, particularly because the survey indicated that women frequently searched for information to understand whether their

pain experiences were normal. However, this would function as a safety and informational feature rather than the application's primary purpose. Finally, I considered implementing a simple feedback mechanism where users could indicate whether a suggested technique helped them. This could potentially improve personalization over time without requiring users to maintain detailed records or complete lengthy tracking activities.

Design Synthesis

The overall synthesis of the formative research was that menstrual pain management involves much more than selecting a relief technique. Participants described a combination of physical relief attempts, emotional coping, preparation, and adjustments to everyday responsibilities. Their approaches varied depending on their previous experiences, the severity of their symptoms, and the resources they had available. While some participants had established routines for managing their pain, these strategies were not necessarily effective in every situation. The research suggested that accessible and personalized support, along with the freedom to respond to pain, could be just as important as the techniques themselves.

Another important insight was that participants already had existing systems for finding menstrual health information and preparing for their periods. The survey suggested that women generally knew which sources they preferred, with search engines and social media being more commonly used than AI agents or chatbots. Although respondents did not express particularly strong levels of trust in the information they found, they were already familiar with these sources and continued using them. This made me reconsider whether introducing another educational resource would meaningfully improve their experience. Instead, I began exploring how a design intervention could complement the resources and routines users already have, without introducing additional work or cognitive burden.

Changes in the Design Idea

At the beginning of the project, my idea was to develop an educational application that would serve as an easily accessible database of menstrual health information. I initially imagined a platform where women could learn about period symptoms, identify relief techniques, and access reliable information whenever they needed it. However, the survey and interviews made me question whether this approach would address the challenges participants were experiencing. Many participants already knew where to find information, but knowing about relief techniques did not necessarily mean they could use them when they were in pain. Their circumstances, responsibilities, emotional states, and access to resources frequently influenced their ability to manage their symptoms.

As a result, my project gradually shifted from an educational application to a more personalized support intervention for women experiencing menstrual pain. Instead of focusing on menstrual cycle tracking or providing a large collection of generic recommendations, I wanted to create something that users could turn to during moments of discomfort. The application would aim to support both the physical and emotional aspects of pain management by helping users access remedies they already find helpful, prepare for situations where pain may occur, and receive

encouragement from people they trust. Potential areas of support included personalized pain-management strategies, gentle movement or exercises, dietary suggestions, and emotional well-being. These would be explored as optional resources rather than presented as guaranteed methods of relieving pain.

The research also introduced an interesting design consideration regarding whether the final intervention should be entirely digital or include a physical component. Since several participants already carried period supplies or maintained personal preparedness routines, I began considering how a physical object could provide convenient access to a digital support system. However, the interviews did not establish a direct need for a new physical product. Therefore, this became an idea to explore during ideation rather than an assumption about what users wanted.

Ideation and Design Features

To explore possible interventions, I conducted a Crazy 8s ideation exercise based on the design implications identified during the research synthesis. The exercise helped me move beyond the original educational application concept and consider different ways to support women during moments of menstrual pain. Although the formative research generated several possible design directions, I prioritized the implications that appeared most relevant to recurring patterns in the survey and interviews. In particular, I focused on personalization, preparedness, situational accessibility, and emotional support. This led me to develop a concept centered around three main features.

The first feature is a **Period Kit List**, which allows users to maintain a digital list of the menstrual products and other essentials they usually carry. The feature would also include optional reminders to replenish items, potentially connected to messaging platforms such as WhatsApp or Messages. This idea was inspired by the preparedness behaviors observed during the interviews, where participants described maintaining their own systems for carrying menstrual supplies and preferred relief items. Rather than replacing those systems, the feature would help users organize and maintain them. It could be particularly useful for people who spend significant time outside their homes and want to ensure that their essential supplies remain accessible.

The second feature is a **Women's Support Community**, designed to provide emotional encouragement during menstrual pain. The interviews highlighted that emotional comfort, acknowledgment, and feeling supported could make painful days more manageable, even when they did not directly reduce physical pain. Building on this finding, I envisioned a private support space where users could connect with trusted friends or close contacts who could send encouraging messages, digital flowers, and small gestures of care. The purpose of this feature would not be to provide medical advice or create another social media platform, but to make emotional support more accessible. Users would have control over whom they invite into their support space and when they want to receive encouragement.

The third feature focuses on **Personalized Remedies Based on the User's Situation and Available Resources**. Rather than recommending generic relief techniques, the application would allow users to identify strategies that have worked for them in the past and suggest options based on their current circumstances. For example, if someone experiences menstrual pain while attending class or commuting, the application could prioritize techniques that are practical in those environments and consider the items they already have in their period kit. This concept directly responds to the research finding that participants' ability to manage pain was often limited by their responsibilities and surroundings. I also considered incorporating a simple feedback mechanism that allows users to indicate whether a technique was helpful, so that future suggestions could better reflect their preferences without requiring extensive tracking.

Physical and Digital Integration
To make the support system more accessible, I also explored the possibility of integrating the application with a physical plush keychain. The concept involves a small plush accessory that users could attach to their backpacks, bags, or keychains. An NFC tag embedded inside the plush would allow users to tap it with a compatible smartphone and open the web application. This could provide a simple entry point to their personalized remedies, period kit list, and emotional support features without requiring them to manually search for the application during moments of discomfort.

The plush keychain was intended to serve as both a convenient access point and a comforting physical companion. Since the interviews suggested that emotional comfort and supportive environments play a role in how participants cope with menstrual pain, I wanted to explore whether a soft, familiar object could contribute to that experience. However, this physical component remains a design hypothesis that would need to be tested with potential users. Further evaluation would help determine whether a plush keychain genuinely improves accessibility and comfort or creates an unnecessary additional product.

Overall Design Direction

Through the formative research, synthesis, and ideation process, my project evolved from an information-focused application into a personalized menstrual pain support system. The main goal is no longer simply to educate users about menstrual pain, but to help them navigate painful moments using resources and strategies that fit their individual circumstances. The proposed intervention combines practical preparedness, personalized relief suggestions, and emotional support while avoiding the expectation that users must continuously track their menstrual cycles or adopt an entirely new routine.

Moving forward, I plan to evaluate whether these proposed features meaningfully address the needs identified during the research. In particular, I want to explore whether users would find value in a digital period kit list, how comfortable they would feel involving trusted people in their pain-management experience, and whether a physical keychain would make accessing support easier. These concepts are still early design directions rather than validated solutions. The next stage of the project will focus on prototyping and gathering user feedback to determine which

features provide meaningful support without adding unnecessary complexity to the experience of managing menstrual pain.

Next Steps

- **Week 1**, I will finalize the core prototype flow, build the first clickable prototype using Lovable or Figma Make, prepare usability testing materials, and recruit participants. I will also start testing ideas for the plush keychain with the NFC tag embedded in it.
- **Week 2**, I will conduct prototype testing, analyze participant feedback, and submit the November 13 milestone.
- **Week 3**, I will refine the prototype based on findings and begin developing a higher-fidelity version of the design alongside early drafts for the final proposal and video storyboard.
- **Week 4**, I will conduct quick follow-up validation, finalize the prototype candidate, and create draft versions of the poster, proposal, and presentation materials.
- **Final Sprint**, I will focus on polishing all deliverables, recording and editing the short video, finalizing the poster and written proposal, rehearsing the stakeholder demo, and preparing the complete final submission package.



Resources

Interview Guide: Shrina Tyarla - Interview Guideline Document
Survey: Reimagining Period Pain Management Interventions - Google Forms.pdf
Affinity Notes: Link for Affinity Notes